Champions League: How well can game performance factors predict number of goals?

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Why it's important

- Critical decision for club management/tactics to increase competitiveness
- Scoring goals is the primary objective in soccer
- Focus on team play
- Suitable for comparing team performance and outcomes
- Design attacking strategies
- Include in data set teams with different playing styles, offensive and defensive strengths

Sample Data

- Sample data was taken from Champions League
- 36 European soccer teams
- Teams from top competitive leagues
- Cross sectional data set from previous season
- Performance variables considered initially:
  - Goals, Attempts on Target, Passes completed, Possession, Saves, Tackles won and Cards

	Wins	Goals	Attempts	Passes	Possession (%)	Tackles won	Saves	Cards
AVG	3,31	13,53	39,72	3267,17	50,01	50,58	27,33	15,36
Min	0,00	5,00	22,00	2115,00	38,10	30,00	14,00	7,00
Max	8,00	23,00	61,00	4911,00	64,00	93,00	49,00	22,00
STD	1,83	5,13	11,42	735,90	5,98	12,39	9,61	4,12



Independent/Dependent Variables

Dependent variable- Goals scored
Independent variables- Attempts on Target and Possession %

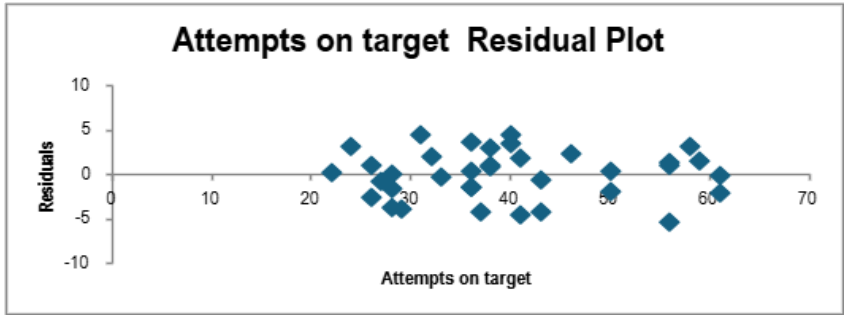
Initial selection variables: Knowing that goals are representative of wins, we compared Goals with Attempts on Target, Passes completed, Possession, Saves, Tackles won and Cards to identify correlation. The process involved running a regression, removing the variable with the highest P-value, then re-running it until we were left with only two variables: Attempts on Target and Possession (%)

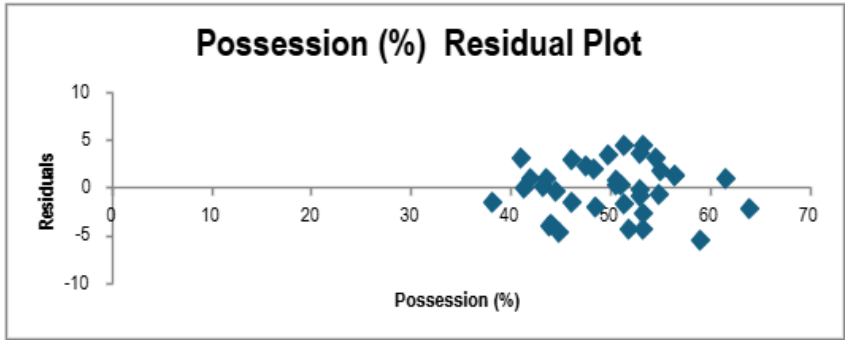
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-8,0379343	6,193710701	-1,2977575	0,20496055	-20,725176	4,64930692	-20,725176	4,64930692
Wins	1,39921118	0,395307424	3,539552	0,00142216	0,58946063	2,20896173	0,58946063	2,20896173
Attempts on target	0,15687364	0,06208072	2,52693005	0,0174362	0,02970705	0,28404023	0,02970705	0,28404023
Passes completed	-0,0006631	0,00144715	-0,4581858	0,65035527	-0,0036274	0,00230129	-0,0036274	0,00230129
Possession (%)	0,25490032	0,184977412	1,37800784	0,17911462	-0,1240087	0,63380938	-0,1240087	0,63380938
Tackles won	0,00377303	0,035901727	0,10509339	0,91705096	-0,0697683	0,07731439	-0,0697683	0,07731439
Saves	0,09217898	0,047594039	1,93677571	0,06292327	-0,005313	0,18967095	-0,005313	0,18967095
Cards	-0,1681356	0,102768597	-1,6360597	0,11302019	-0,3786475	0,04237636	-0,3786475	0,04237636



Regression

- Linear trend line applied
- No discernible pattern in Residual plot. Assume a linear Regression.
- Multivariate with backward elimination
- The model is highly significant and explains over 70% of the variation in goals scored





	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	665,6513055	332,825653	43,0174178	6,41E-10
Residual	33	255,3209167	7,73699748		
Total	35	920,9722222			

- **Data Points:** 36
- **R^2 =** 0.723
- **Adjusted R^2 =** 0.706
- **Standard Error:** 2.78
- **F-statistic =** 43.02
- **p-value =** 6.41×10^{-10}

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-10,56286275	4,185447092	-2,5237119	0,01660748	-19,078219	-2,0475066	-19,078219	-2,0475066
Attempts on target	0,276152558	0,055480711	4,97745164	1,9714E-05	0,1632762	0,38902891	0,1632762	0,38902891
Possession (%)	0,262366642	0,105998056	2,47520239	0,0186241	0,04671198	0,47802131	0,04671198	0,47802131



Implication of analysis

Variables show:

- Attempts on target:
  - Direct measure of offensive effectiveness
  - Expected to have a more positive impact on goals
  - Past season Champions League data suggests that with every ~ third attempt a goal is scored.
- Possession (%)
  - Higher possession may lead to more scoring opportunities
  - The chance to score increase by 0.25% if 1% in possession is additionally gained.
- Intercept
  - Even though the intercept is significant it can be dismissed as it is not explainable in football terms (0% possession is unlikely). The data points range around higher positive values and the regression line is negatively extrapolated to 0.



Recommendation:

- Teams should focus on:
  - Generating shots on target, which statistically lead to more goal and therefore wins
  - Using possession productively, not passively
- Possession alone is not enough unless it leads to real chances, for instance Man city has a higher ball possession than Liverpool, however Liverpool has more goals scored than Man city.

